GiteHubLink:- https://github.com/Siddharth-Tapkir/CNL

PART 1 — BUILD THE TOPOLOGY

You already have the topology shown in your screenshot, so if you've already placed everything, **don't delete anything**.

You need:

1 × Switch

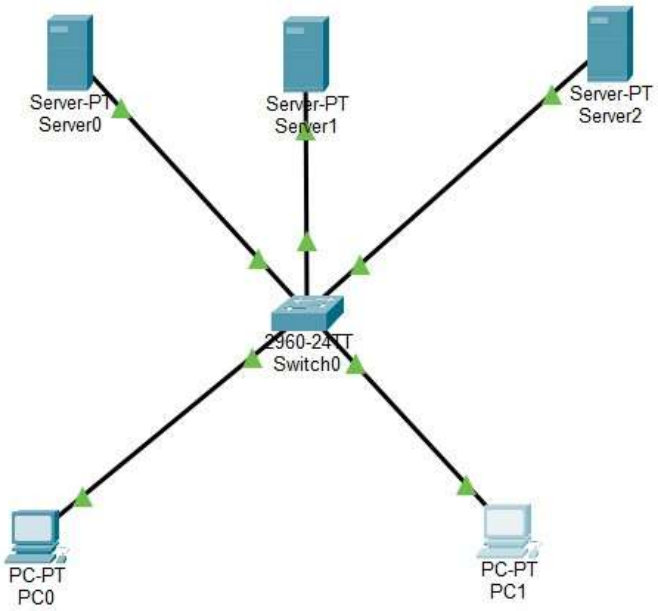
Network Devices → Switches → 2960

3 × Servers

End Devices → Server

2 × PCs

End Devices → PC



connecons:

Server0 → Switch0

Server0 FastEthernet0

↓

Switch0 Fa0/1

Server1 → Switch0

Server1 FastEthernet0

↓

Switch0 Fa0/2

Server2 → Switch0

Server2 FastEthernet0

↓

Switch0 Fa0/3

PC0 → Switch0

PC0 FastEthernet0

↓

Switch0 Fa0/4

PC1 → Switch0

PC1 FastEthernet0

↓

Switch0 Fa0/5

PART 2 — IP ADDRESS PLAN

Since we have **three different servers**, each server gets its own stac IP.

Let's use:

Device IP Address Role

Server0 192.168.1.2 DHCP

Server1 192.168.1.3 DNS

Server2 192.168.1.4 Email

PC0 DHCP Client

PC1 DHCP Client

Subnet: 255.255.255.0

We'll use:

Network: 192.168.1.0/24

PART 3 — SERVER0 = DHCP SERVER

Click:

Server0 → Desktop → IP Configuraon

Select:

Stac

Enter:

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

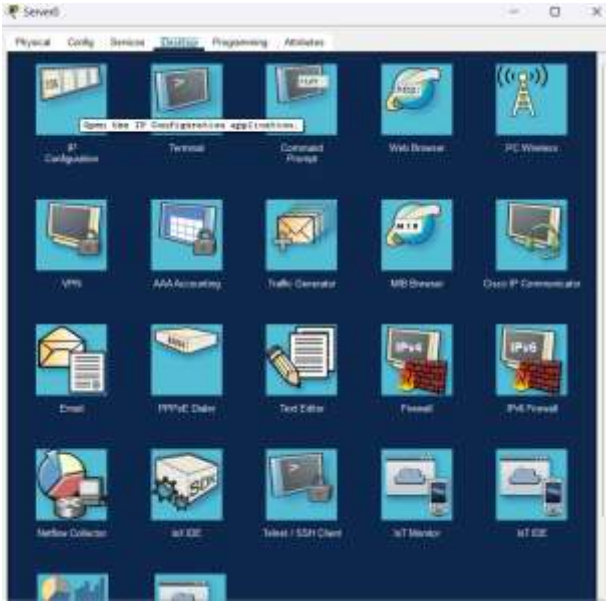
DNS Server: 192.168.1.3

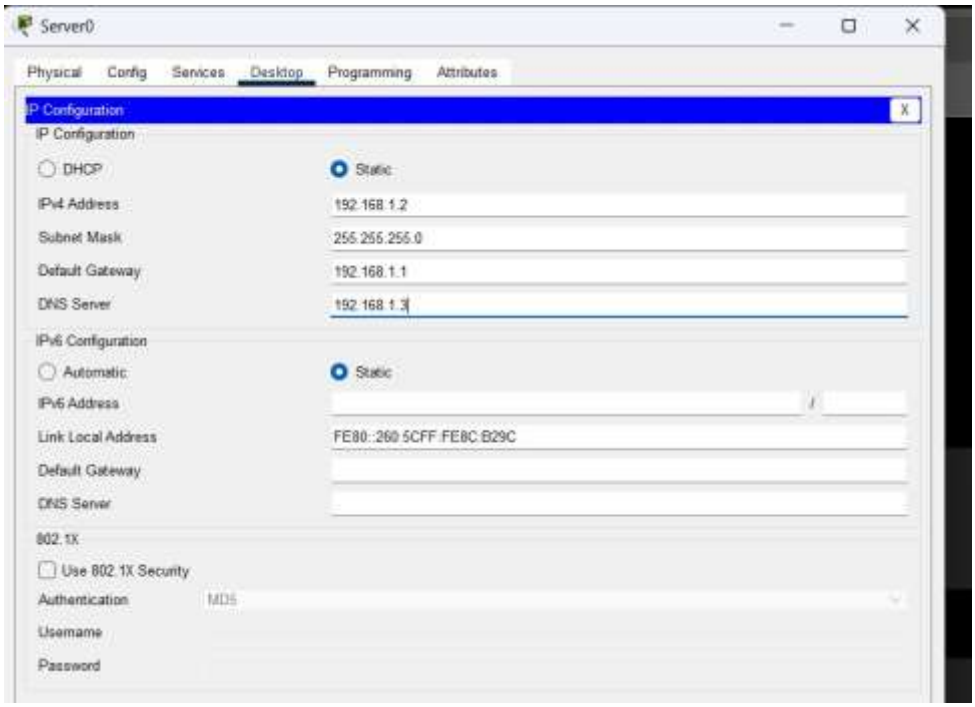
So Server0 becomes:

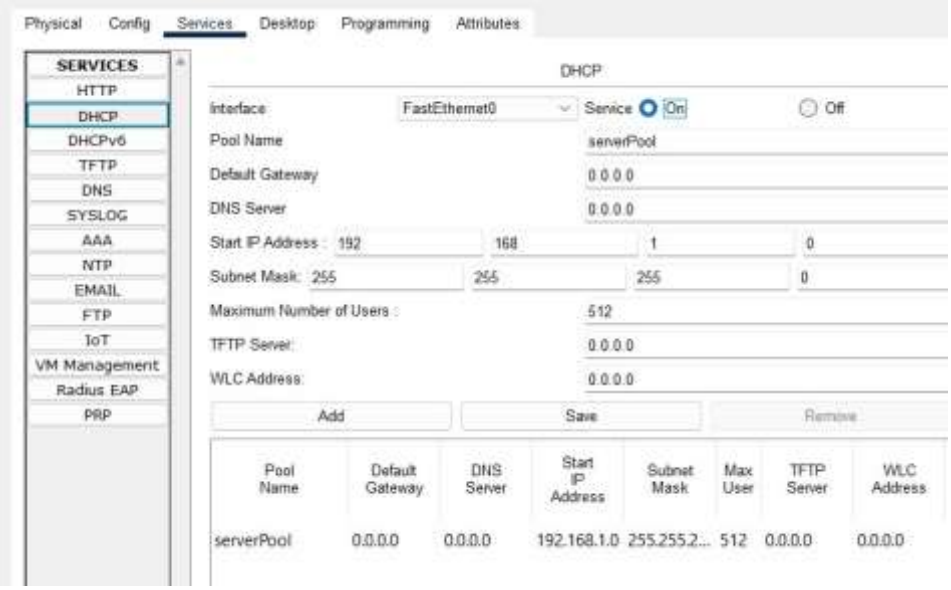
Server0

DHCP

192.168.1.2




Server0
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.1
DNS Server
192.168.1.3
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:5CFF:FE8C:B29C
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password



Part 4 — TURN ON DHCP Go:

Server0 → Services → DHCP

Set:

DHCP: ON

Create the pool.

Pool Name

Pool1

Default Gateway

192.168.1.1

DNS Server

This is VERY important.

Because **Server1 is our DNS server**, enter:

192.168.1.3

Starng IP Address

192.168.1.10

Subnet Mask

255.255.255.0

Maximum Users

20

So:

DHCP: ON

Pool Name: Pool1

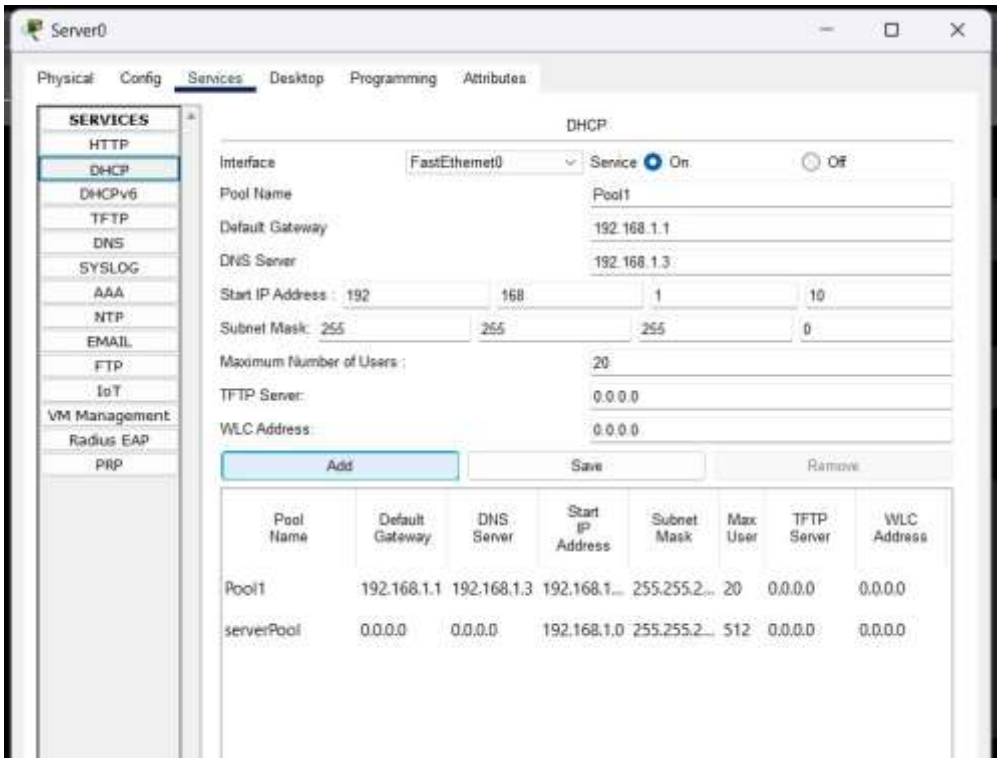
Default Gateway: 192.168.1.1

DNS Server: 192.168.1.3

Starng IP: 192.168.1.10

Subnet Mask: 255.255.255.0

Maximum Users: 20



Part 5 — SERVER1 = DNS SERVER

Now click **Server1**.

Go:

Desktop → IP Configuraon

Select:

Stac

Enter:

IP Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

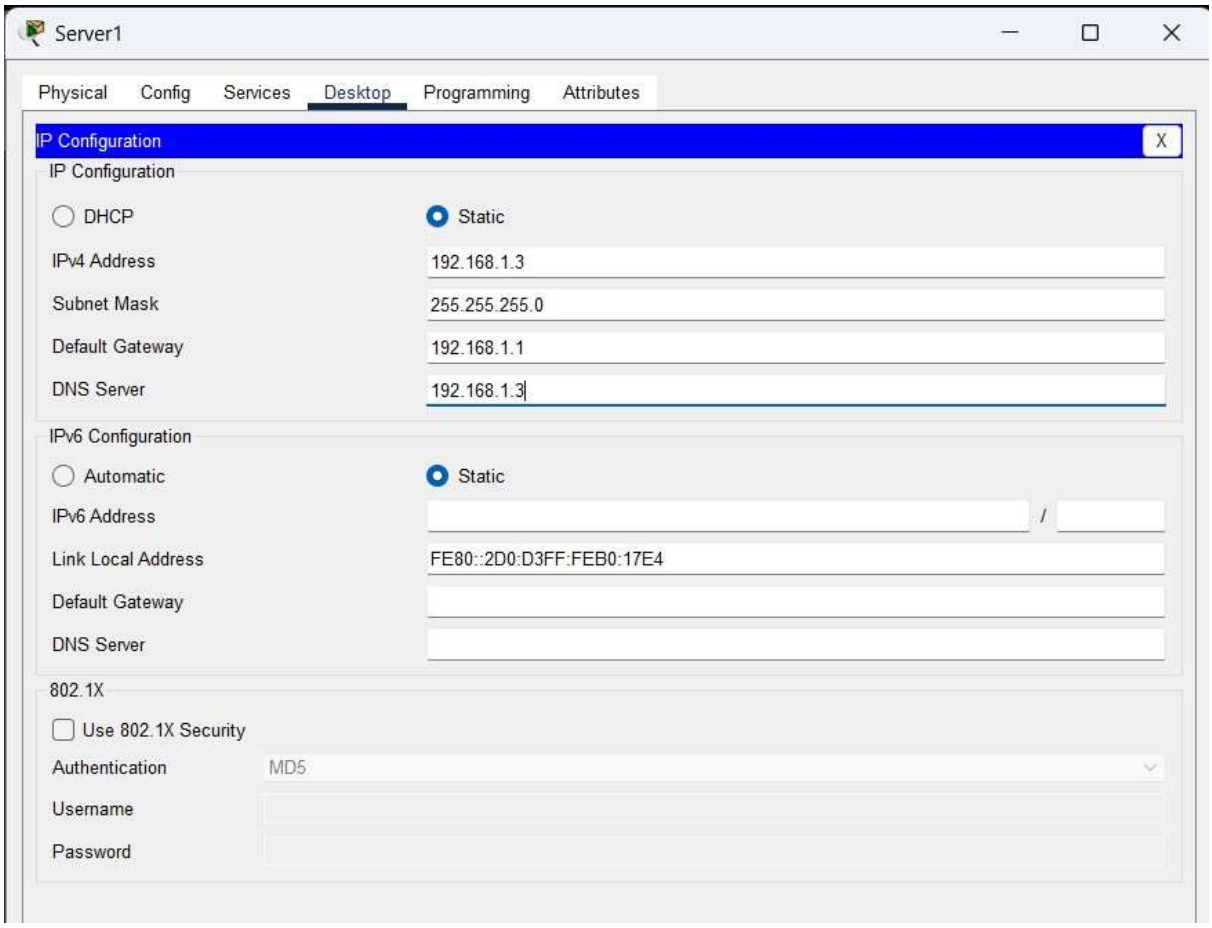
DNS Server: 192.168.1.3

So:

Server1

DNS

192.168.1.3



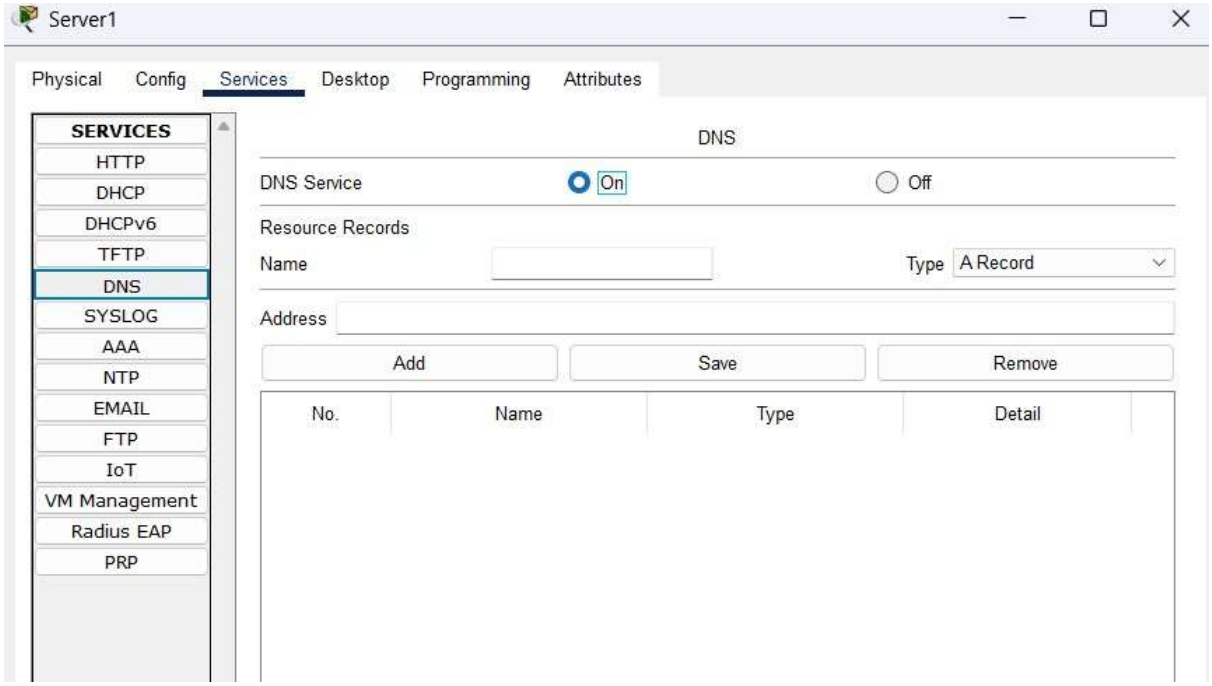
Part 6 — TURN ON DNS

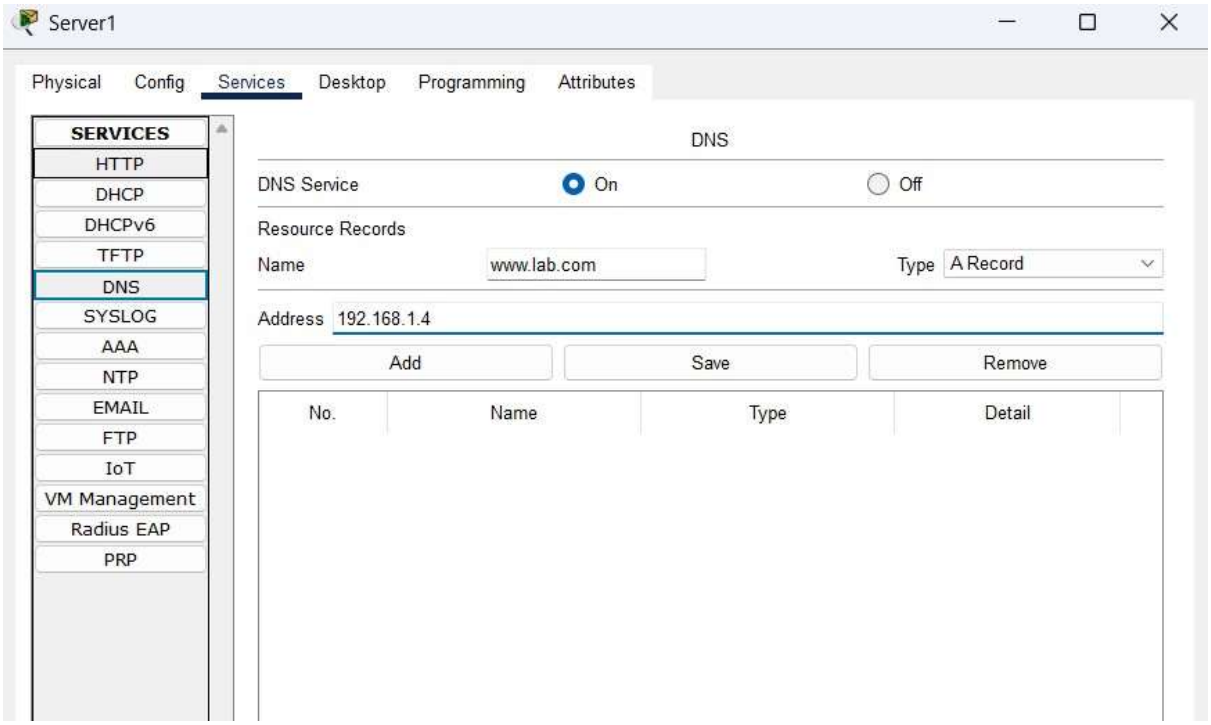
Go:

Server1 → Services → DNS

Set:

DNS: ON





Part 7 — SERVER2 = GMAIL SERVER

Now click **Server2**.

Go:

Desktop → IP Configuraon

Select:

Stac

Enter:

IP Address: 192.168.1.4

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

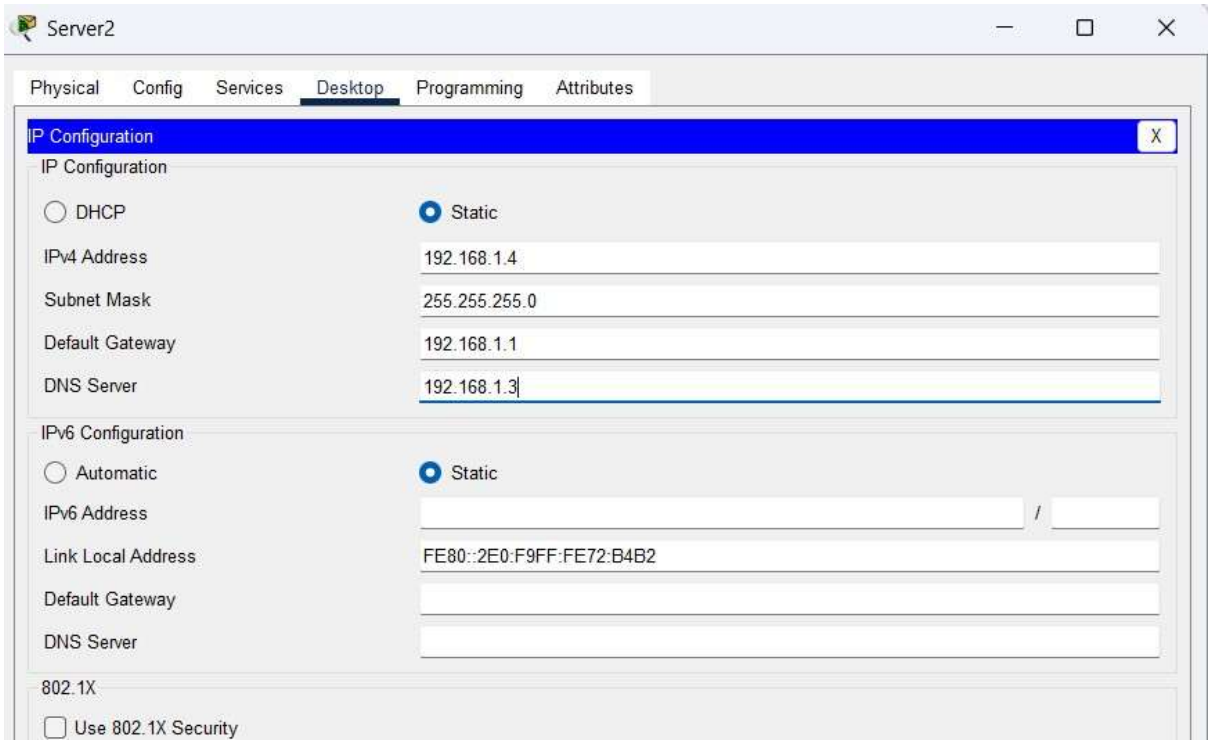
DNS Server: 192.168.1.3

So:

Server2

GMAIL / EMAIL

192.168.1.4



Part 8 — CONFIGURE EMAIL SERVER

Go:

Server2 → Services → EMAIL

Turn:

SMTP: ON

POP3: ON Set

the domain:

lab.com

Now create two users.

User 1

Username: alice

Password: 1234

Click **Add**.

User 2

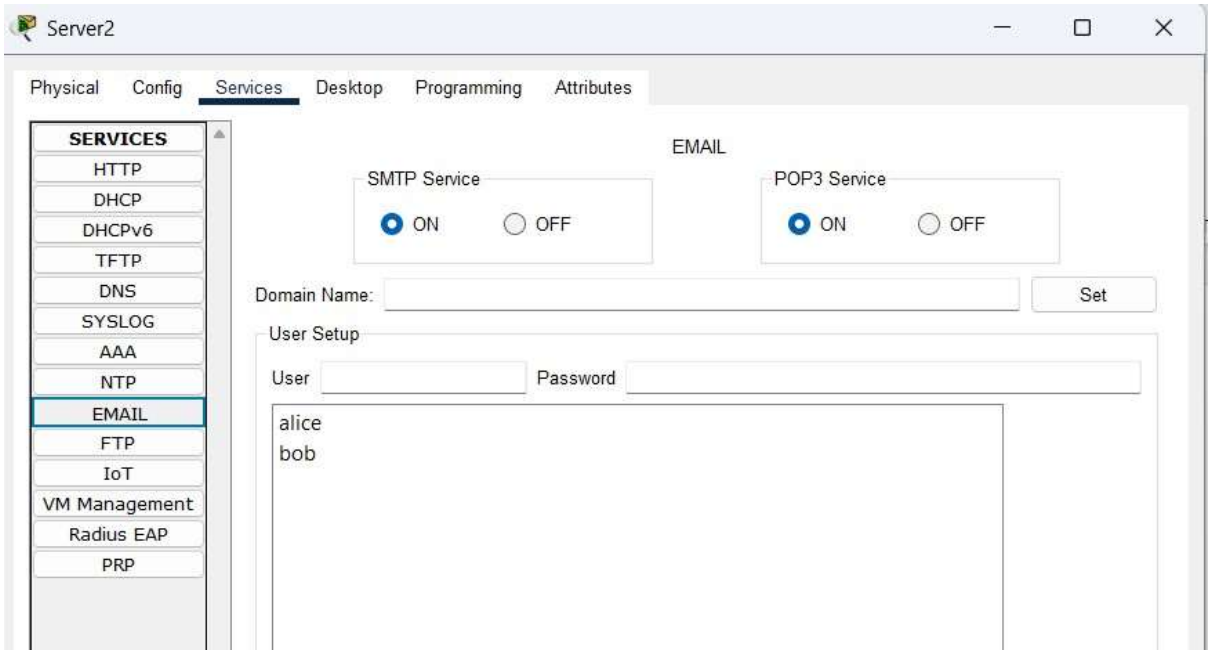
Username: bob

Password: 1234 Click

Add.

Now we have: alice@lab.com

bob@lab.com



PART 9 — CONFIGURE PC0 AND PC1 USING DHCP

The two PCs are DHCP clients. They should obtain their IP addresses automacally from Server0, which is configured as the DHCP server.

PC0:

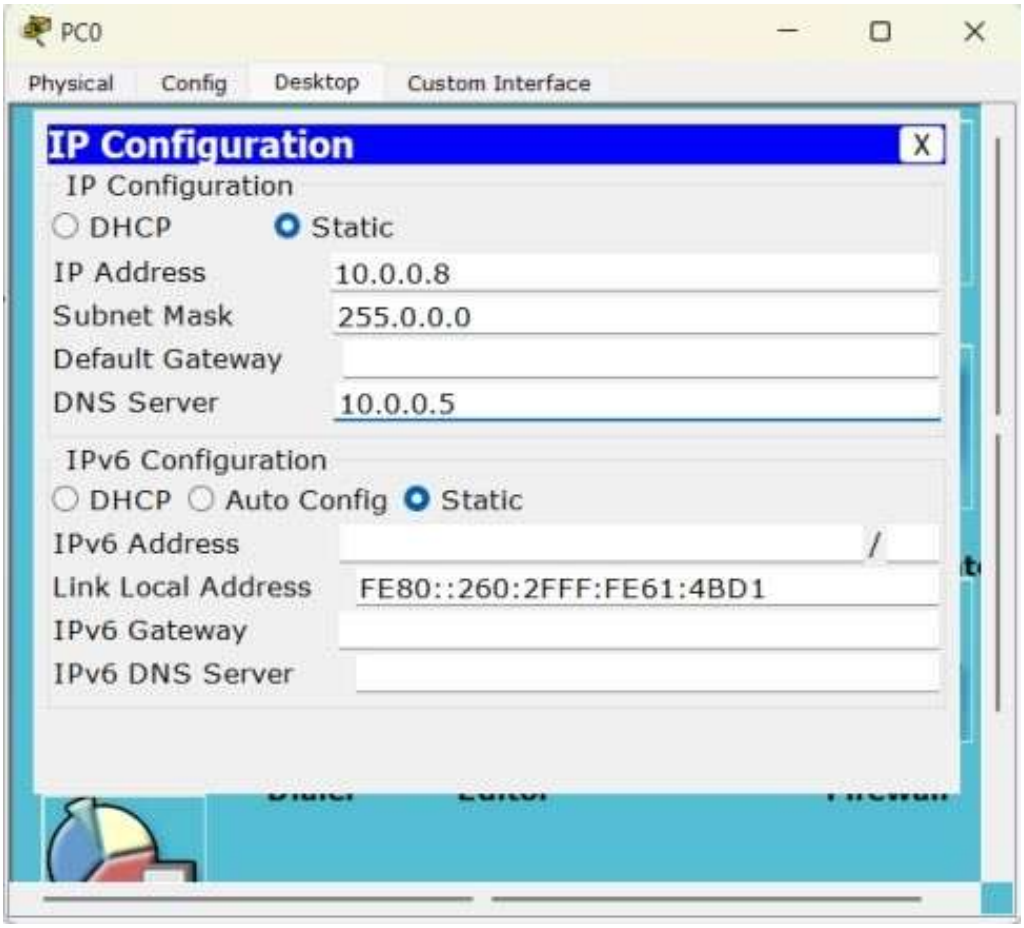
- Open PC0 → Desktop → IP Configuraon.
- Select DHCP.
- Wait for the address to be assigned automacally.
- Expected address: 192.168.1.10.
- Expected subnet mask: 255.255.255.0.
- Expected DNS server: 192.168.1.3.

PC1:

- Open PC1 → Desktop → IP Configuraon.
- Select DHCP.
- Wait for the address to be assigned automacally.

- Expected address: 192.168.1.11.
- Expected subnet mask: 255.255.255.0.
- Expected DNS server: 192.168.1.3.

NOTE: The exact DHCP-assigned address can change if the pool has already been used. With a fresh configuraon, PC0 should normally receive 192.168.1.10 and PC1 should receive 192.168.1.11.



PART 10 — CONFIGURE EMAIL CLIENTS

Server2 is the email server at 192.168.1.4. SMTP and POP3 are ON, and the domain configured in this experiment is lab.com.

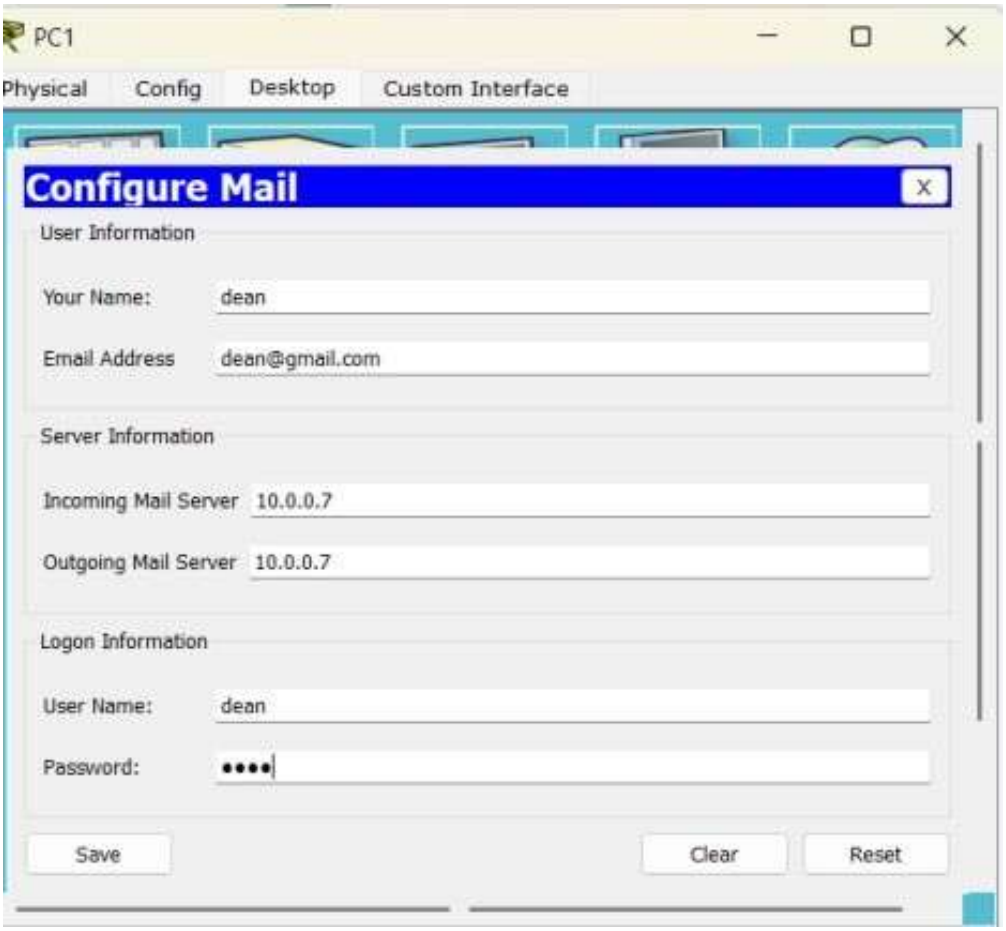
Configure PC0 for Alice:

- Open PC0 → Desktop → Email → Configure Mail.
- Your Name: Alice
- Email Address: alice@lab.com
- Incoming Mail Server: 192.168.1.4
- Outgoing Mail Server: 192.168.1.4
- Username: alice
- Password: 1234

- Click Save.

Configure PC1 for Bob:

- Open PC1 → Desktop → Email → Configure Mail.
- Your Name: Bob
- Email Address: bob@lab.com
- Incoming Mail Server: 192.168.1.4
- Outgoing Mail Server: 192.168.1.4
- Username: bob • Password: 1234
- Click Save.



NOTE: The reference screenshot shows the same Packet Tracer Mail Browser/Configure Mail workflow, but its sample values belong to the supplied answer booklet. Enter alice@lab.com and bob@lab.com with mail server 192.168.1.4 in this experiment.

PART 11 — SEND AN EMAIL

Aer both email clients are configured, send a test message from Alice (PC0) to Bob (PC1).

1. Open PC0 → Desktop → Email → Compose.

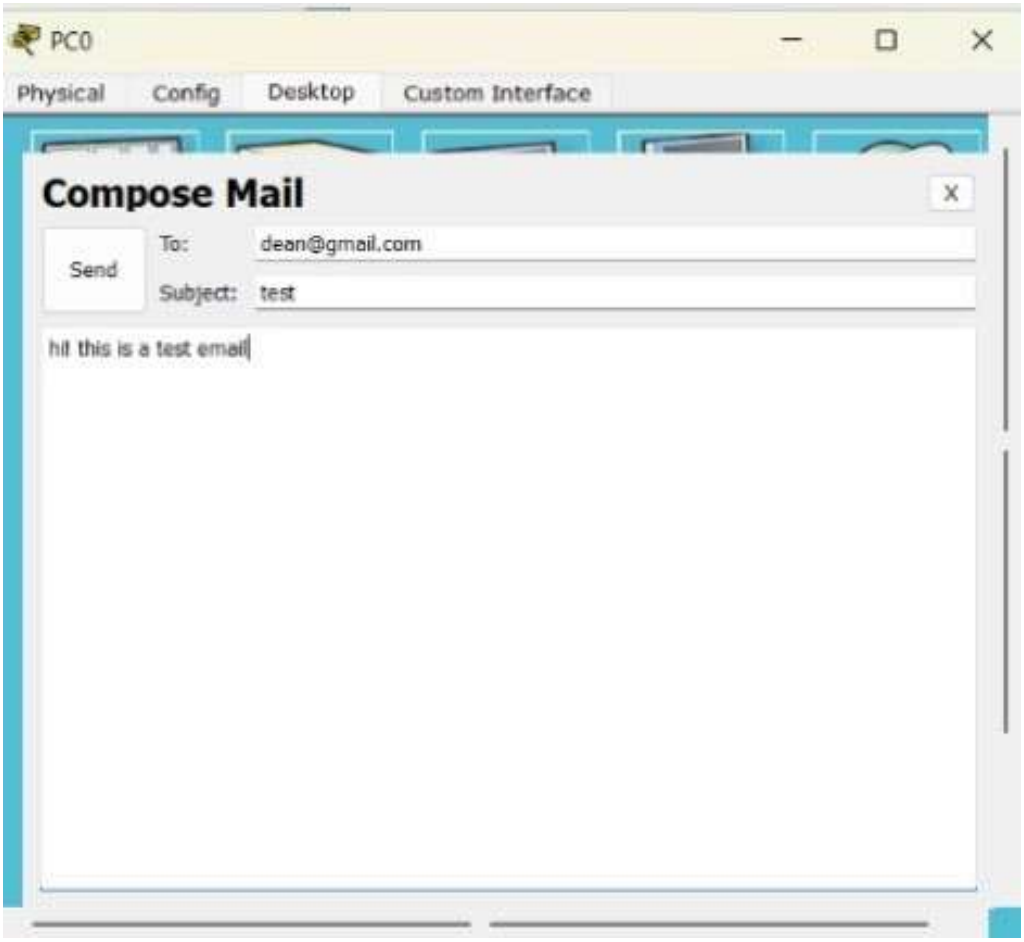
2. To: bob@lab.com.

3. Subject: Test.

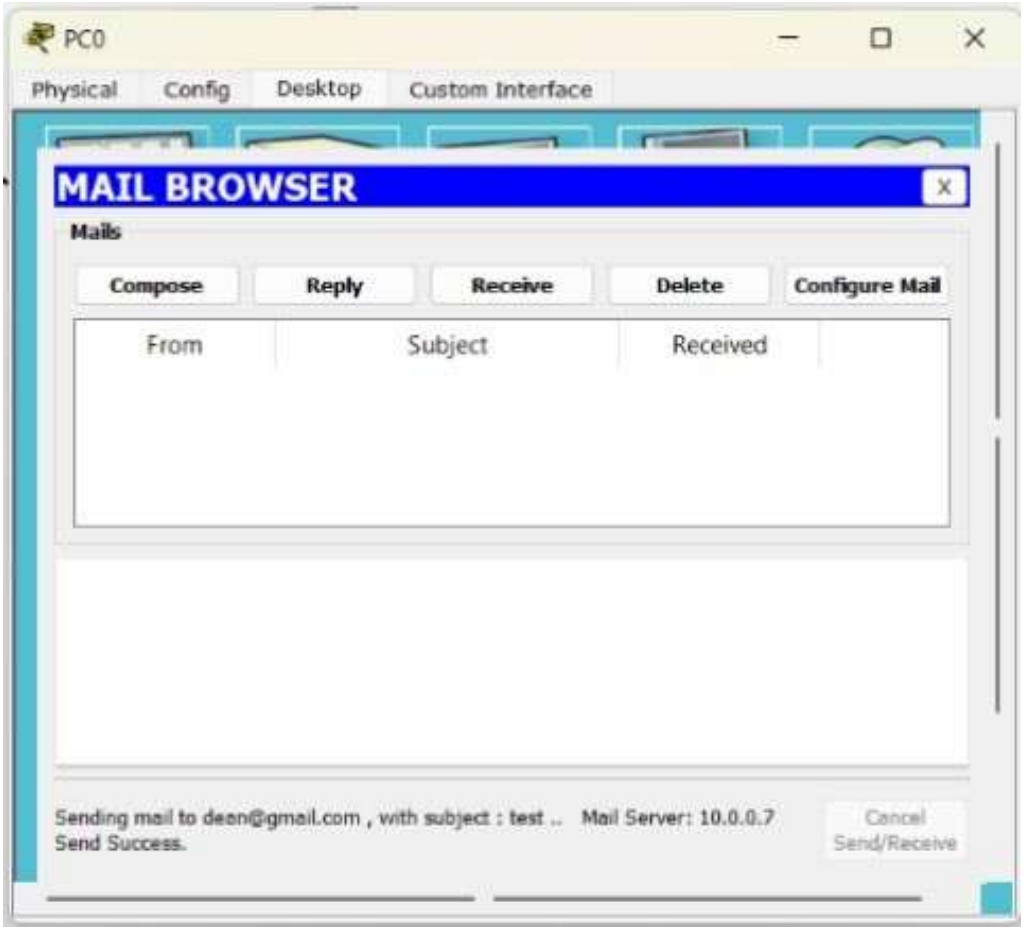
4. Message: Hi Bob, this is a test email.

5. Click Send.

6. The Mail Browser should report that the mail was sent successfully.



Reference screenshot showing the Compose Mail window.



Reference screenshot showing successful mail transmission.

NOTE: In the reference screenshots, the example uses dean@gmail.com and the old 10.0.0.7 mail server. For this experiment, the correct desnaon is bob@lab.com and the mail server is 192.168.1.4.

PART 12 — RECEIVE AND READ THE EMAIL

Now open Bob's mailbox on PC1 and retrieve the message sent by Alice.

1. Open PC1 → Desktop → Email → Mail Browser.

2. Click Receive.

3. Wait for the message to appear in the mail list.

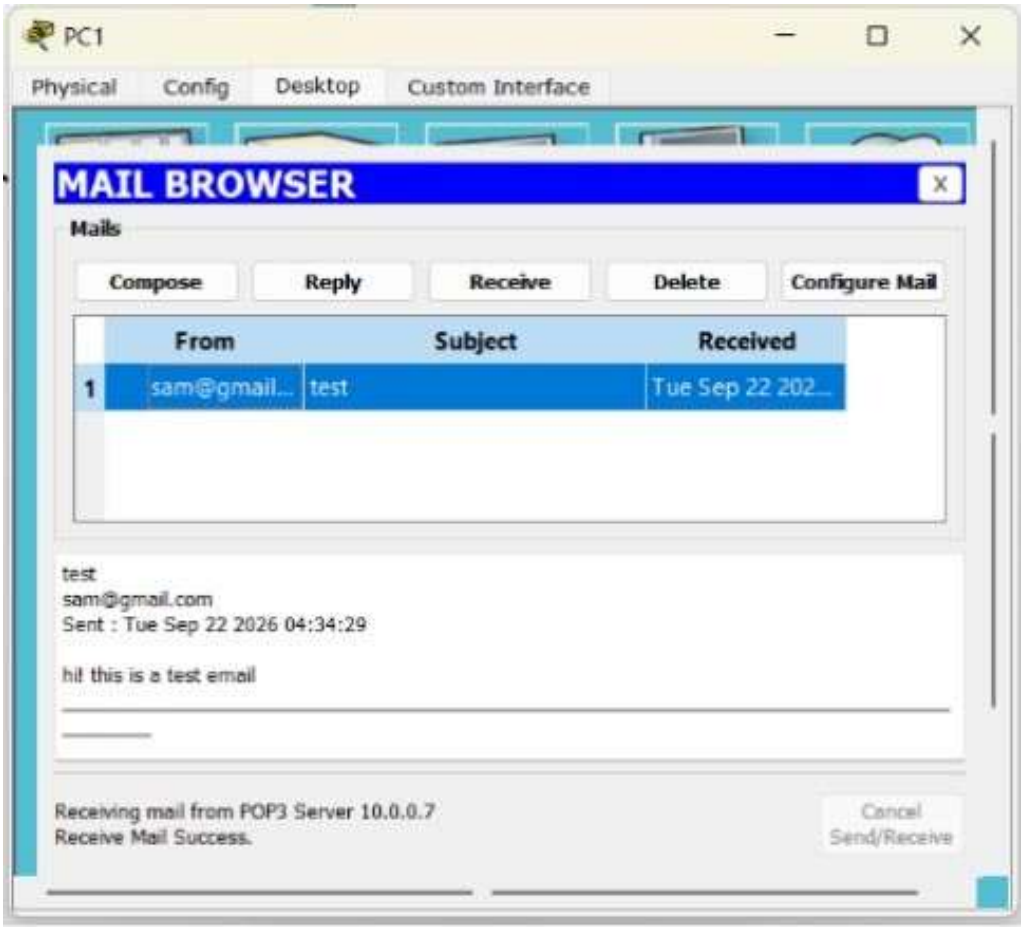
4. Select the received message.

5. Verify that the sender is alice@lab.com and the subject is Test.

6. Open the message and verify its contents.



Reference screenshot showing a received message in the Mail Browser.



Reference screenshot showing the contents of the received email.

PART 13 — REPLY TO THE EMAIL

Finally, reply from Bob (PC1) back to Alice (PC0) to verify two-way email communicaon.

1. Open the received message on PC1.

2. Click Reply.

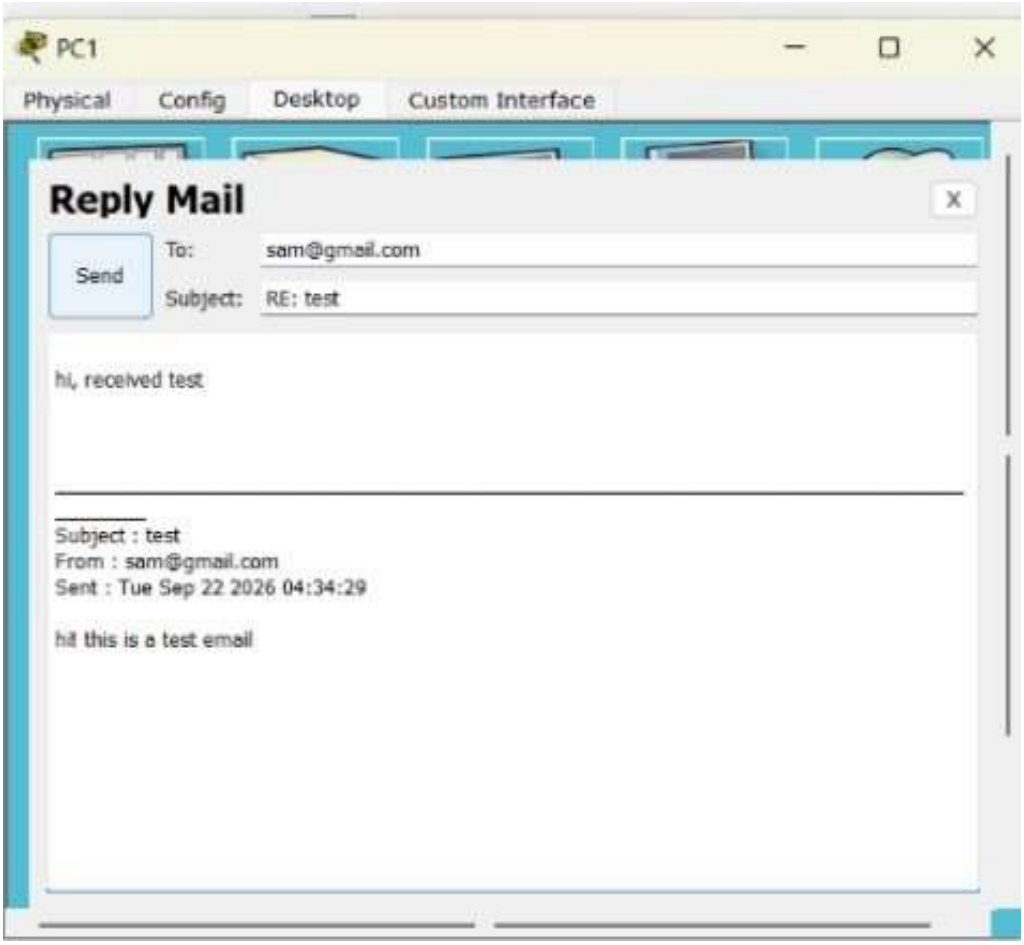
3. The recipient should be alice@lab.com.

4. Enter a reply such as: Hi Alice, I received your test email.

5. Click Send.

6. Open PC0 → Desktop → Email and click Receive.

7. Verify that the reply from bob@lab.com appears in Alice's mailbox.



PART 14 — VERIFY THE COMPLETE NETWORK

Use Packet Tracer's PC command prompt to verify that DHCP, DNS and the email server are reachable.

1. On PC0 and PC1, open Desktop → Command Prompt.
2. Run ipconfig and confirm that each PC has a 192.168.1.x address, subnet mask 255.255.255.0 and DNS server 192.168.1.3.
3. Run ping 192.168.1.3 to verify connecvity to the DNS server.
4. Run ping 192.168.1.4 to verify connecvity to the email server.
5. If a DNS resource record has been added, test it with ping <domain-name>.
6. Use the Email applicaon to send, receive and reply to a message. Successful delivery confirms that the applicaon-layer service is working.

Expected final configuraon

Device	Address	Role	Expected status
Server0	192.168.1.2	DHCP	DHCP service ON
Server1	192.168.1.3	DNS	DNS service ON
Server2	192.168.1.4	Email	SMTP + POP3 ON
PC0	DHCP → 192.168.1.10*	Alice client	Can send/receive
PC1	DHCP → 192.168.1.11*	Bob client	Can send/receive

NOTE: *The exact DHCP lease can differ if the pool has already allocated addresses. The important requirement is that both PCs receive valid 192.168.1.x addresses from Server0.

RESULT

The Packet Tracer network was configured with one switch, three servers and two PCs. Server0 provides DHCP, Server1 provides DNS, and Server2 provides email services using SMTP and POP3. The PCs obtain their IP configuraon through DHCP, and Alice and Bob can exchange email messages through the configured mail server.

CONCLUSION

Thus, DHCP, DNS and email services were successfully configured and tested in Cisco Packet Tracer. Connecvity was verified between the end devices and servers, and successful send, receive and reply operaons demonstrate working email communicaon.